StepWise

MOBILE LEARNING APP · ENGLISH & MATHS · AGES 10–15 · NIGERIA

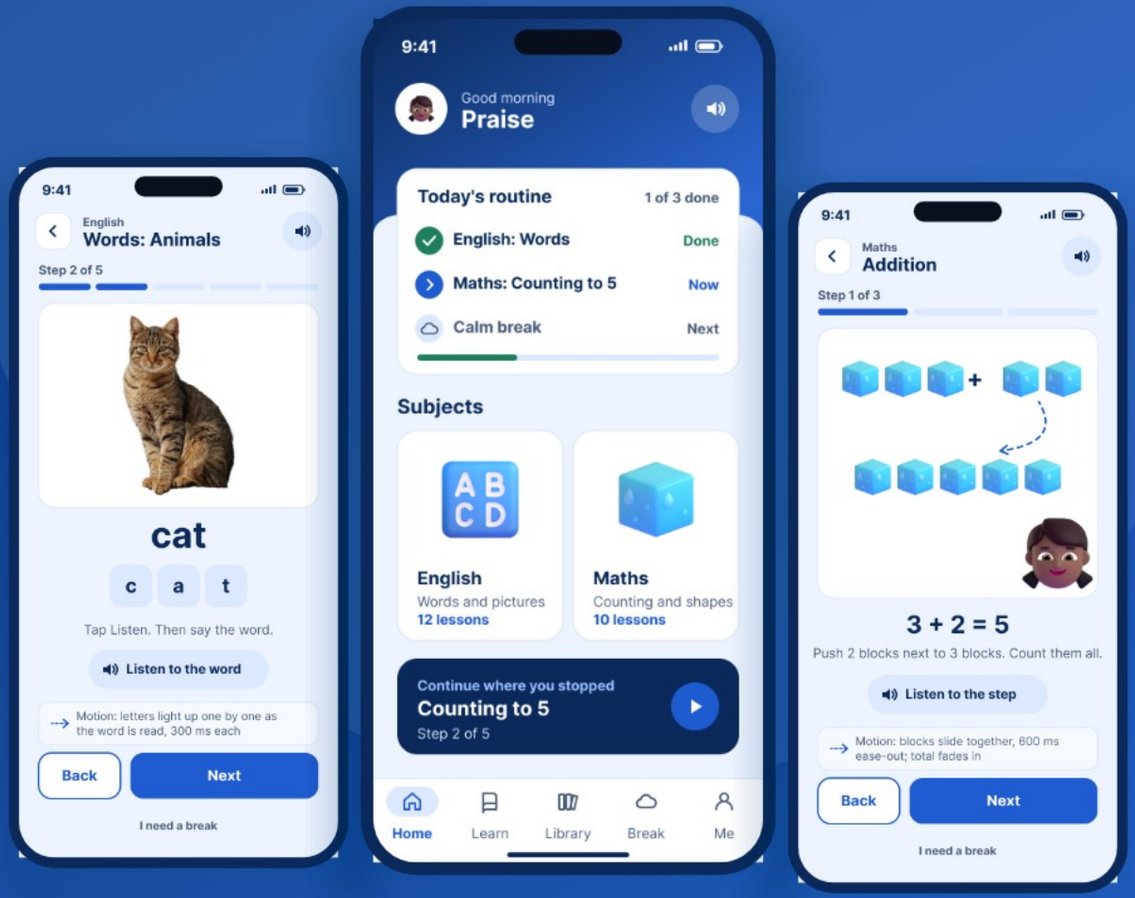
Visual, step-by-step lessons for autistic learners

StepWise teaches one idea per screen, in a predictable routine, with a listen button on every lesson and a break available at any moment. Designed by Clement Olatunde for autistic learners, families and teachers in Nigeria.

Role · UX/UI Designer

Tools · Figma, FigJam

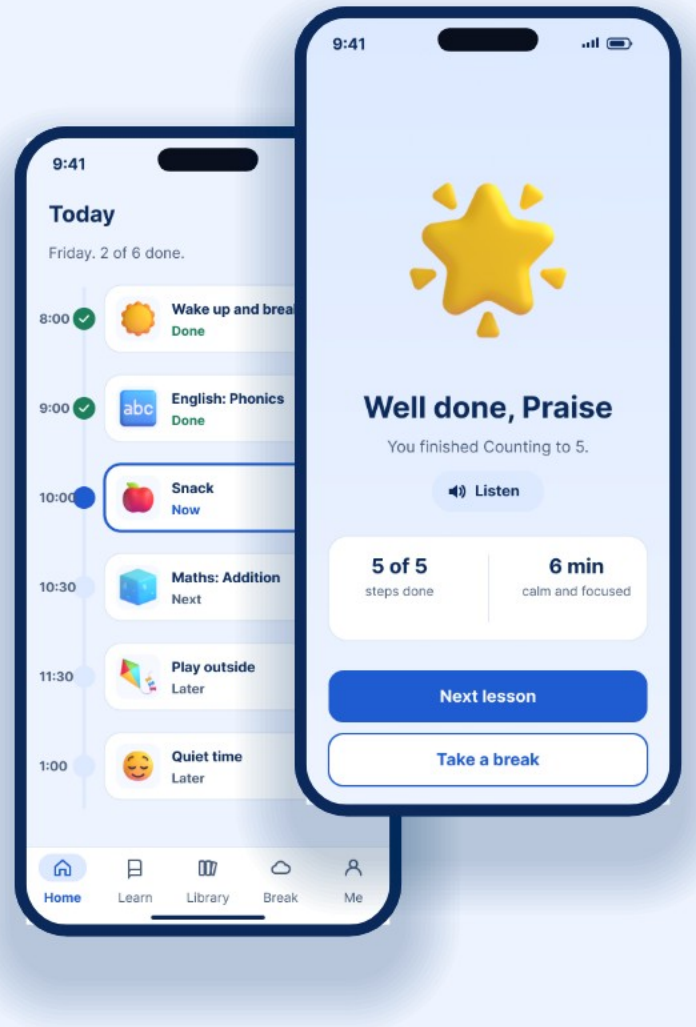
Platform · iOS, 390×844



01 · OVERVIEW

A calm, predictable way to learn English and Maths

StepWise is a mobile learning app for autistic learners aged 10–15 in Nigeria. Every lesson is broken into small visual steps, read aloud on demand, and rewarded calmly. Parents and teachers see progress in a simple dashboard.



ROLE

UX/UI Designer — research synthesis, UX flows, UI, design system

DESIGNER

Clement Olatunde

TIMELINE

October 2026 · concept to high-fidelity screens

TOOLS

Figma, FigJam

PLATFORM

iOS · 390×844 frames · Inter

DELIVERABLES

30 screens, cover, design system, illustration credits

The problem

Most learning apps are busy, loud and unpredictable: dense text, fast animations, sudden sounds and many choices per screen. For many autistic learners that is a barrier before the lesson even starts. In Nigeria, where specialists are few and families pay for most care, parents and teachers need tools they can use themselves.

The approach

Design the whole app around structure: one idea per screen, a visible routine, the same layout on every lesson, audio for every instruction, calm feedback and a break that is always one tap away.

The result

A 30-screen iOS design for English (phonics, words, sentences, stories) and Maths (counting, + − × ÷, brackets, fractions), with a progress system, learning library, calm break and a parent/teacher dashboard.

Mainstream learning apps are designed to excite. Many autistic learners need them to be predictable.

In Nigeria, autistic children are often diagnosed late, and parents and teachers carry most of the day-to-day support. Published guidance on designing for autistic people and people with cognitive disabilities is consistent about what gets in the way; most popular learning apps do the opposite.

01

Sensory overload

Bright, high-contrast palettes, confetti, autoplaying sound and constant movement can be distressing rather than motivating.

Source: WHO autism fact sheet ("unusual reactions to sensations"); W3C COGA: "Limit interruptions", "Avoid too much content".

02

Unpredictable layouts

Each screen looks different, buttons move, and pop-ups interrupt. Without a visible routine the learner cannot know what comes next.

Source: W3C COGA: "Use a consistent visual design"; TEACCH Tip #1: "Visual schedule".

03

Walls of text and idioms

Instructions are long, abstract and full of figures of speech; reading support is hidden behind settings.

Source: W3C COGA: "Use clear words", "Use literal language", "Keep text succinct", "Separate each instruction".

04

No way out

When a learner is overwhelmed the only options are to push on or quit. There is rarely a calm, sanctioned pause built into the flow.

Source: W3C COGA, "Cognitive stress": "I need to avoid and recover from mental fatigue".

Why this matters in Nigeria

The WHO estimates that in 2021 about 1 in 127 people worldwide were autistic, but notes that prevalence in many low- and middle-income countries is unknown; a 2019 review of Nigerian research calls for a nationwide prevalence study. Nigerian studies report late diagnosis, stigma, too few trained professionals, poor access to care and costs paid out of pocket. In a Lagos survey of 230 parents of autistic children, only 1.3% had good knowledge of autism and 53% reported stigma from others. A learning app cannot replace therapy, but it can give parents and teachers a calm, structured tool for home or class, and remove the barriers the guidance names again and again: noise, clutter, unpredictability and language that is harder than the lesson itself.

Sources: WHO "Autism" fact sheet, 2025; Bakare et al., 2019; Oduyemi et al., BMC Psychology, 2021 (full list in section 16).

Six rules every screen had to obey

These principles were set before any screen was drawn, from the guidance in the research section. Each one is checked against every one of the 30 screens.



One idea per screen

A lesson step shows one word, one sum or one picture. Nothing competes for attention; the next idea is one tap away.



A predictable routine

The same header, the same step bar, the same button positions on every lesson. Today's schedule is always visible from Home.



Listen, everywhere

Every instruction, word and question has a large listen button. Reading is supported, never assumed.



Show the steps

"Step 2 of 5" and a segmented progress bar tell the learner exactly where they are and how much is left.



Calm rewards

A soft 3D star, a gentle "Well done" and a tally of badges. No confetti, no flashing, no sudden sounds.

A break is always allowed

"I need a break" sits on every lesson and leads to a breathing exercise. Stopping is a feature, not a failure.

04 · RESEARCH INSIGHTS (SECONDARY RESEARCH)

What published research and guidance say, and what they changed in the design

No primary user research has been run yet; the planned usability test is described below. These insights come from published research on autism in Nigeria and from international accessibility and autism-education guidance, each with its source. The right column shows the decision it led to in StepWise.

INSIGHT	SOURCE	DESIGN DECISION IN STEPWISE
Autism in Nigeria is often diagnosed late. Reported reasons: stigma, low awareness, too few trained professionals, poor access to care and costs paid out of pocket. Involving parents and non-specialists in support is recommended.	Bakare, Taiwo, Bello-Mojeed & Munir, "Autism Spectrum Disorders in Nigeria: A Scoping Review…", J. Health Care for the Poor and Underserved, 2019.	A parent and teacher side that needs no specialist training: plain-language settings, a progress dashboard, and lessons an adult can run at home or in class.
In a Lagos survey of 230 parents of autistic children, only 1.3% had good knowledge of autism; 53% reported stigma from others and 83.5% had internalised it.	Oduyemi et al., "Internalization of stigma among parents of children with autism spectrum disorder in Nigeria", BMC Psychology, 2021.	Adult screens in plain words with no clinical jargon; progress stays with the family and teacher, with no leaderboards or public sharing.
Children with disabilities face barriers to education from discrimination, stigma and exclusion in school; inclusive education is the most effective way for all children to learn.	UNICEF, "Children with disabilities" (inclusive education).	A teacher role at sign-up and a teacher dashboard, so a mainstream class teacher like Mrs Bello can support autistic learners in her class.
Across low- and middle-income countries, affordability (mainly of phones) is the top reported barrier to getting online, and the cost of data and devices limits use once online.	GSMA, The State of Mobile Internet Connectivity 2025: "Affordability of Internet-Enabled Handsets and Data".	Phone-first screens (390 × 844); still images and short read-aloud clips instead of video; nothing plays on its own.
Use clear words and literal language, keep text succinct, use a consistent visual design, make each step clear, limit interruptions and avoid too much content.	W3C, Making Content Usable for People with Cognitive and Learning Disabilities (COGA), W3C Group Note, 2021.	A blue-only palette on soft backgrounds, 14–36 px Inter, one short instruction per screen, descriptive buttons ("Listen", "Next step", "I need a break").
Visual schedules show what is done and what comes next, stay in a consistent place and help prepare for transitions: "structure you can see".	UNC TEACCH Autism Program, "TEACCH Tip #1: Visual Schedule" (Structured TEACCHing).	A Daily schedule screen, a visual step bar and "Step 2 of 5" on every lesson, the same lesson layout across subjects, and a predictable end state ("Well done").
Autistic people may react unusually to sensations, and their abilities and needs vary and can change over time.	World Health Organization, "Autism" fact sheet, updated 17 Sep 2025.	No autoplay audio, no flashing or confetti, calm reward animation only, "I need a break" → breathing exercise on every lesson; sound, text size and motion switches in Settings.
Offer information in more than one form (text, audio, image) and let learners act in more than one way.	CAST, Universal Design for Learning Guidelines 3.0, 2024 (Representation; Action & Expression).	Listen button on every instruction; answers by tapping pictures or typing/voice (Comment screen); real photos for animals, 3D objects for maths.
Segment content into learner-paced parts and keep extraneous material out (segmenting and coherence principles).	R. E. Mayer, Multimedia Learning, 2nd ed., Cambridge University Press, 2009.	One idea per screen with a Next button, no decorative clutter around the lesson object.
Contrast ≥ 4.5:1 for text (1.4.3); targets ≥ 24×24 CSS px (2.5.8); let users stop moving content (2.2.2); avoid animation that is not essential (2.3.3).	W3C, Web Content Accessibility Guidelines (WCAG) 2.2, 2023.	Measured text contrast 4.6:1 to 12.7:1 on the palette, all ≥ 4.5:1; 44–56 px tap targets; reduce-motion and text-size switches in Settings.

PLANNED USABILITY TEST (NOT YET RUN)

Moderated sessions in Lagos with 5–6 autistic learners aged 10–15 and their parent or teacher, in a familiar setting (home or school), one subject per session. Tasks: open today's schedule, complete one English step and one Maths step with the listen button, take a break and return, find today's badge. We will record where the routine breaks down, where audio is used, and how the break is received, then revise the flows before development. No results are claimed in this case study until that test has happened.

Who we are designing for

These are proto-personas: assumptions drawn from the published research and guidance above and the project brief, not from interviews. They will be validated or corrected in the planned usability test.



Praise

PROTO-PERSONA

12 · autistic learner · Lagos, Nigeria

CONTEXT (ASSUMED)

- Learns best from pictures and real photos; reads short words but tires quickly with long text.
- Likes to know the order of the day and gets anxious when an activity changes without warning.
- Enjoys numbers when they are objects she can count; loud praise and surprises make her withdraw.

NEEDS

- Hear any instruction again, as many times as she wants.
- See how many steps are left before the lesson ends.
- Stop and breathe without losing her progress.

FRUSTRATIONS

- Screens that flash, play music or jump between layouts.
- Being asked two things on one screen.
- Time pressure and loud "rewards".



Mrs Bello

PROTO-PERSONA

38 · primary teacher and parent · Lagos

CONTEXT (ASSUMED)

- Supports two autistic learners in a class of 30 and her own son at home; uses a phone more than a laptop.
- Wants to see which topics are done and where a child paused or asked for a break.
- Prepares printable materials and short videos for lessons.

NEEDS

- A dashboard that shows progress per subject, time on task and notes in one glance.
- Control over sound, text size and motion for each child.
- A library of lesson materials she can reuse.

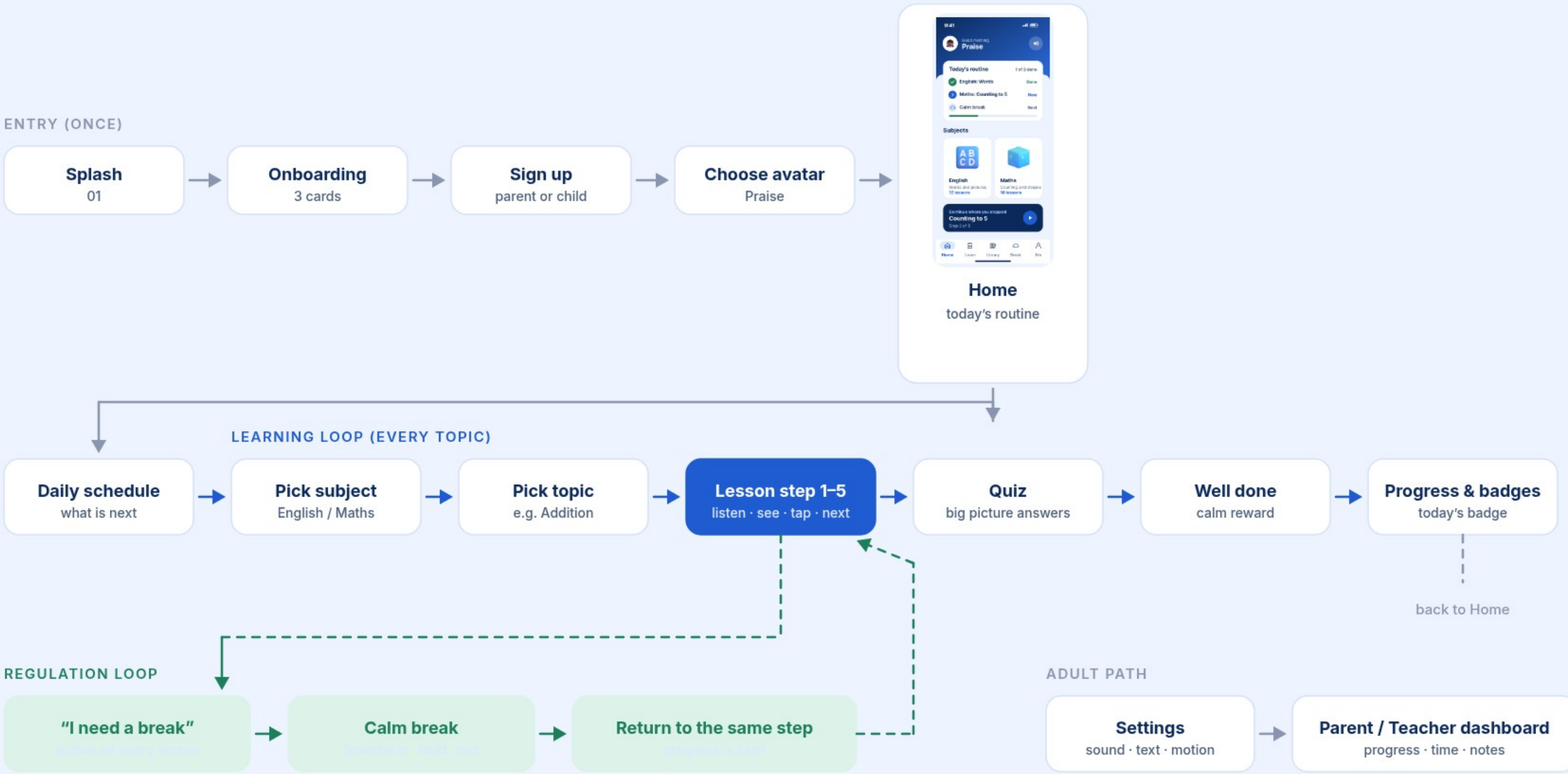
FRUSTRATIONS

- Apps that need an account per child or hide settings deep in menus.
- Gamified streaks that punish a missed day.
- Reports full of charts she cannot act on.

06 · USER JOURNEY & FLOW

Praise’s first session: from opening the app to her first badge

One main path, repeated identically for every topic, plus a break loop that is reachable from any lesson step and always returns to the same step.



Reachable from any lesson step; the step and its progress are kept.

FIRST-SESSION JOURNEY (ASSUMED, TO BE TESTED)

1 Arrive

Opens the app with her teacher; sees today’s routine, not a feed.

“I know what happens next.”

2 Learn

Taps Maths → Addition; hears the instruction; counts blue blocks; taps Next.

“One thing at a time.”

3 Pause

Feels tired at step 3; taps “I need a break”; breathes with the circle.

“I can stop and it waits for me.”

4 Finish

Returns to step 3, finishes the quiz; sees one soft star and her badge.

“I did the whole thing.”

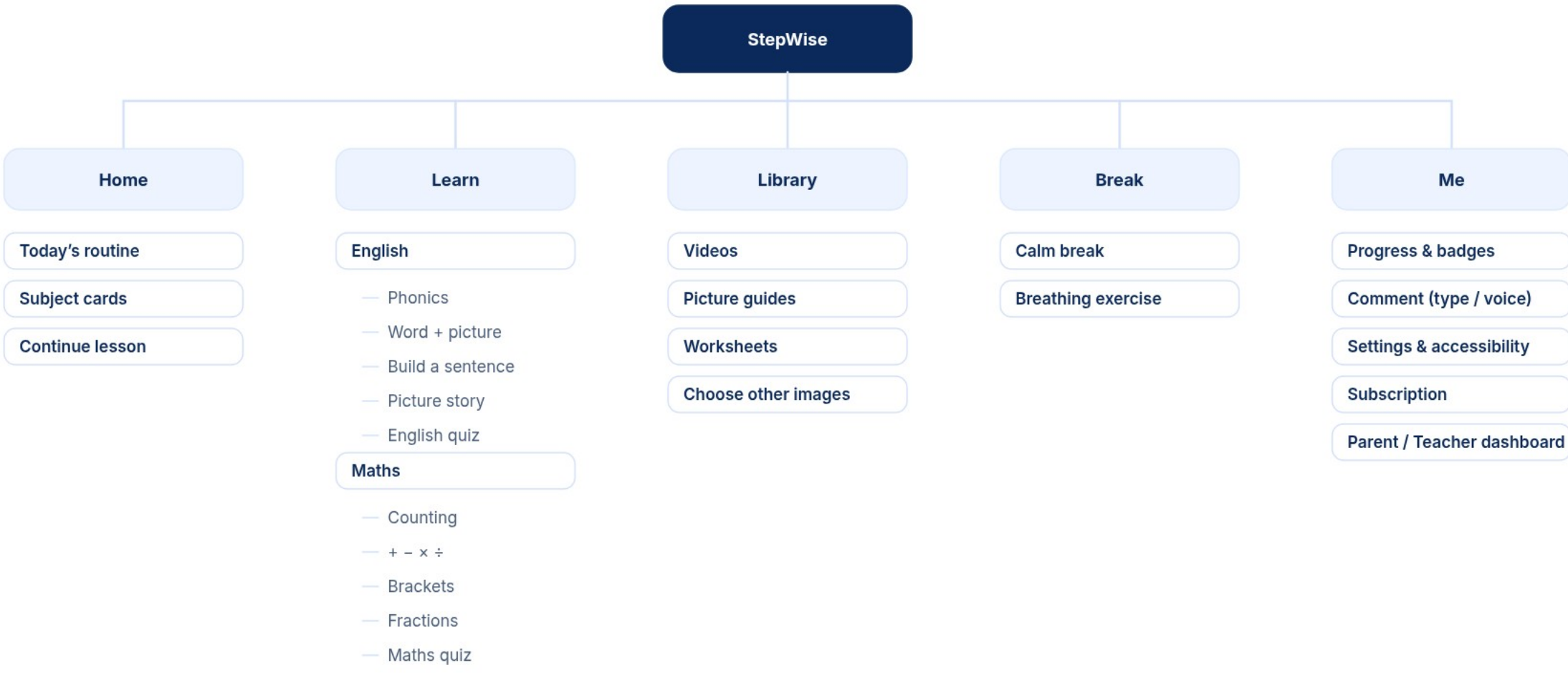
5 Adult check

Mrs Bello opens the dashboard: Addition done, 6 min, one break taken.

“I can see what to do next with her.”

Five tabs, two subjects, one lesson pattern

A flat structure: nothing is more than two taps from Home. Both subjects use the same topic list → lesson step → quiz pattern, so learning one topic teaches the whole app.



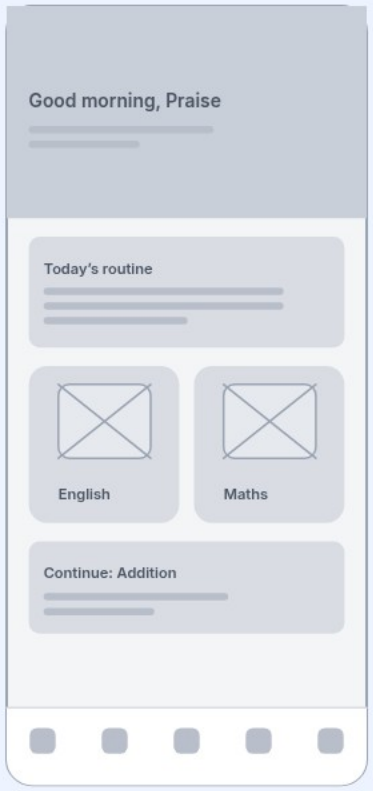
ENTRY, ONCE PER DEVICE

Splash → Onboarding (3 cards) → Sign up (parent or child) → Choose avatar → Home. After that the app always opens on Home, so the routine is the first thing the learner sees.

08 · WIREFRAMES

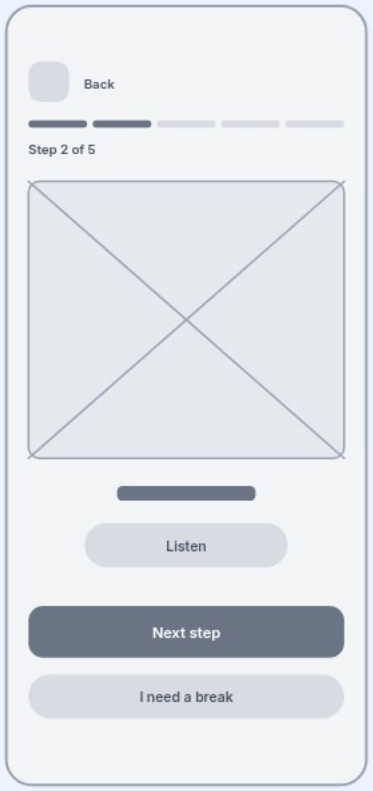
Low-fidelity layouts settled the routine before any colour

Five greyscale wireframes fixed what every screen shares: header with a back button and a step bar, one content object, one listen pill, one primary action and the break button. Colour and 3D objects came later.



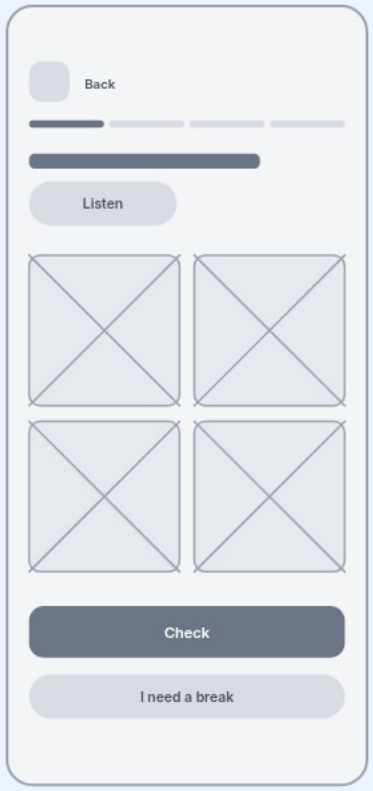
Home

Routine first, two subject cards, continue card.



Lesson step

One object, one word, listen, next, break.



Quiz

Question read aloud, four big picture answers.



Well done

One calm star, two facts, next or break.



Parent dashboard

Progress per subject, time, breaks, notes.

09 · DESIGN SYSTEM

Blue, soft and consistent: the kit behind all 30 screens

A small system on purpose. One typeface, one hue family, two radii, three button types. Everything a learner taps looks the same on every screen.

COLOUR

Navy
#0B2A5B
headings, header gradient start

Primary
#1F5BD1
buttons, links, step done

Primary light
#2F6FE4
header gradient end

Soft blue
#DCE9FF
pills, step to-do, chips

Ice
#EEF4FF
screen background

White
#FFFFFF
cards

Muted text
#4A5D7A
secondary text

Calm green
#1E7F5C
success, break

TYPOGRAPHY · INTER

The cat sits on the mat

Title 36 Bold
Lesson titles, “Well done”

The cat sits on the mat

Heading 28 Bold
Screen headings

The cat sits on the mat

Subheading 24 SemiBold
Card titles, sums

The cat sits on the mat

Body 18 Medium
Instructions, buttons

The cat sits on the mat

Body 16 Regular
Descriptions

The cat sits on the mat

Label 14 Medium
Chips, captions, step count

Minimum 14 px on screen; lesson words 36–120 px. Line length kept under 40 characters for learners.

COMPONENTS

Next step — Primary · 56 px · radius 16

Take a break — Secondary · outline

Listen — Listen pill · 48 px · Back 44 px

Step bar · 5 segments · “Step 2 of 5”

Step 2 of 5

Counting
Blue blocks · 5 steps
Next

Card · white · radius 20 · 1.5 px line

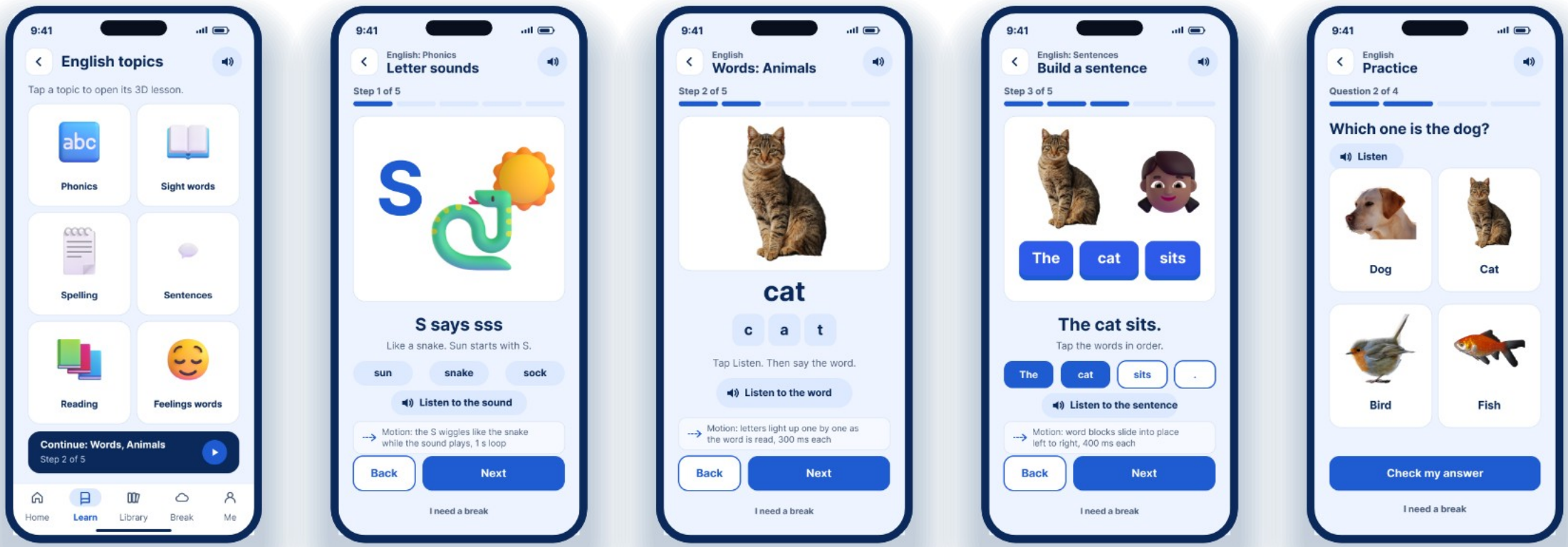
OBJECTS & ICONS · MICROSOFT FLUENT 3D (MIT) + REAL PHOTOS

3D objects are human-made (Microsoft Fluent Emoji, MIT). Animals are real photographs from Wikimedia Commons, cut out. Scene illustrations by Storyset. No AI-generated imagery.

10 · KEY SCREENS — ENGLISH

Phonics to picture stories, one idea at a time

Each English topic uses the same lesson shell. The object changes; the layout, the listen pill and the break button never move.



English topics

Six topics as big cards with a 3D object and a one- or two-word label, plus a “Continue” card that remembers the step the learner stopped at.

Phonics

One letter, one sound, three pictures that start with it (“S says sss”). The listen pill plays the sound; the step bar shows where in the set the learner is.

Word + picture

A real photograph of a cat above the word “cat”, split into its letters. Real photos rather than cartoons, so the word is tied to the real thing.

Build a sentence

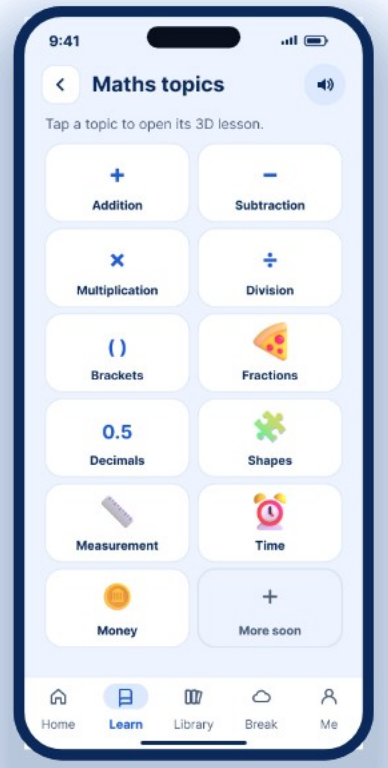
The learner taps the words in order and each lands in its slot. Only the words needed are shown, and the finished sentence is read aloud.

English quiz

The question (“Which one is the dog?”) is read aloud; the four answers are large real photos. A wrong answer is designed to fade softly rather than shake or buzz.

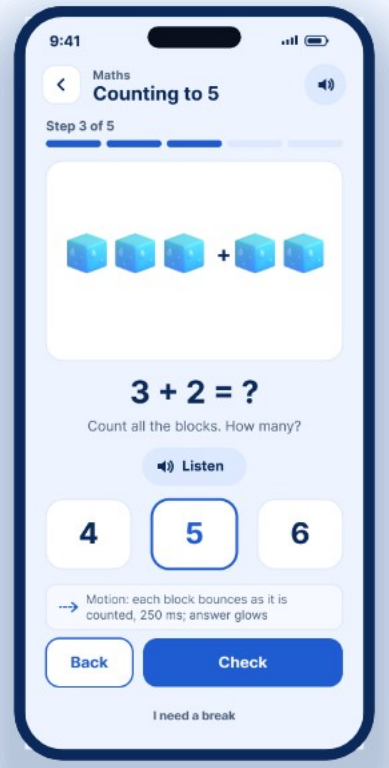
Counting objects before counting numbers

Maths is taught with things the learner can see and count: blue blocks, apples, cookies, pizza slices. The sum is always written as real text beside the objects.



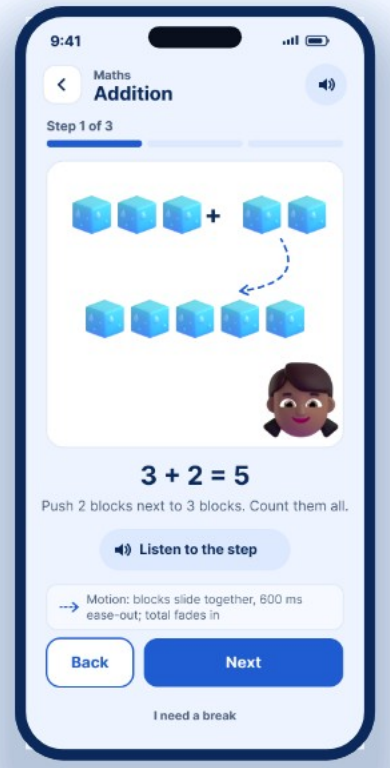
Maths topics

Twelve topics in a fixed grid, each with a symbol and a one-word label. The order never changes, so the path through maths is the same every day.



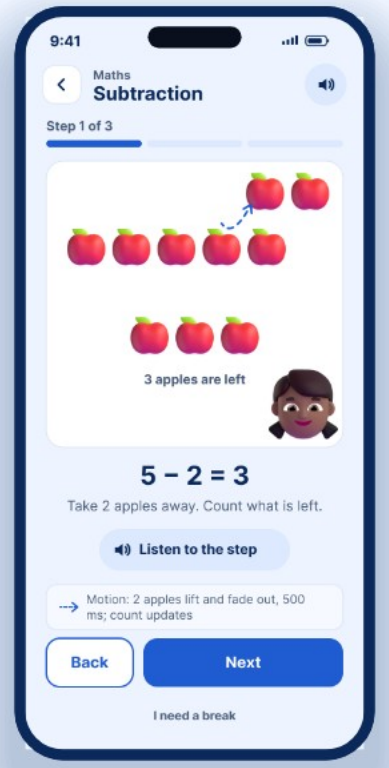
Counting

Blocks are counted one at a time (each bounces gently as it is counted) and the learner picks the total from three large numbers. Nothing moves on its own.



Addition

Two groups of blue blocks slide together into one. The sum 3 + 2 = 5 is real text, read aloud with “Listen to the step”, and the step bar marks progress.



Subtraction

Two apples lift away and fade; the three that are left can be counted on screen. The sum 5 − 2 = 3 is written as text beside the objects.



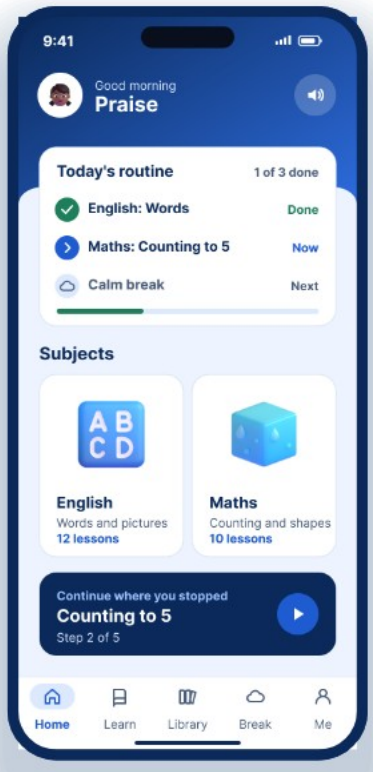
Fractions

Pizza slices make ½, ¼ and ¾ concrete: the coloured slices are the fraction. Each fraction is written as text under the pizza it describes.

12 · KEY SCREENS — ROUTINE, REWARDS & REGULATION

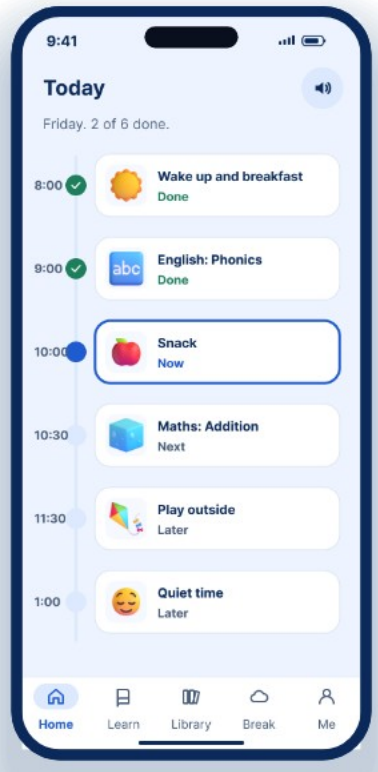
The parts of the app that are not lessons do the most work

Home, the daily schedule, the reward, the calm break and the badge shelf give the day a shape. They are deliberately quiet.



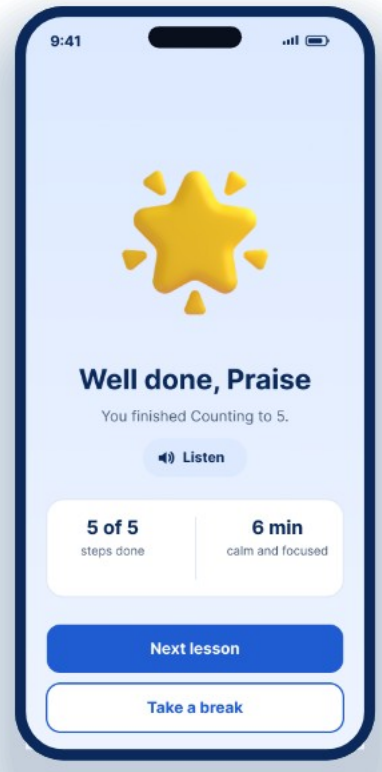
Home

Greets Praise by name, shows today's routine first, then two subject cards and a "continue" card. No feed, no notifications, no streak counter.



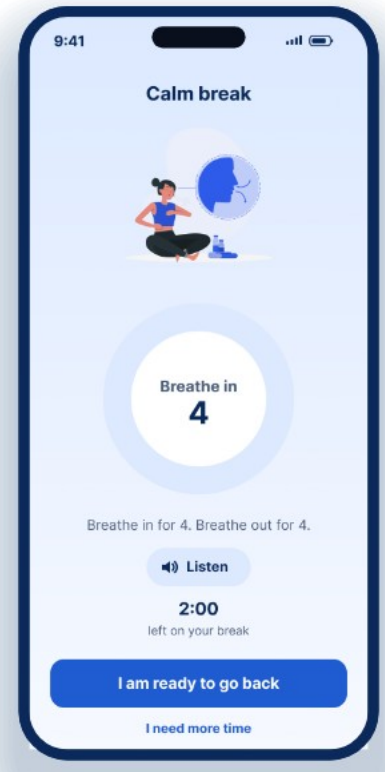
Daily schedule

A visual schedule, as recommended in autism-education guidance: each activity has a picture, a time and a status (done, now, next).



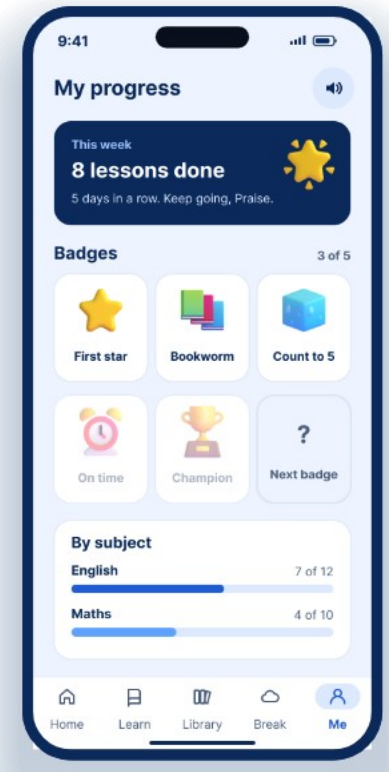
Well done

One soft 3D star, a plain "Well done, Praise", two facts (steps, minutes) and the choice to continue or rest. No confetti, no fanfare.



Calm break

Reached from "I need a break" on any lesson. A breathing circle with slow in / hold / out and a button back to the same step.



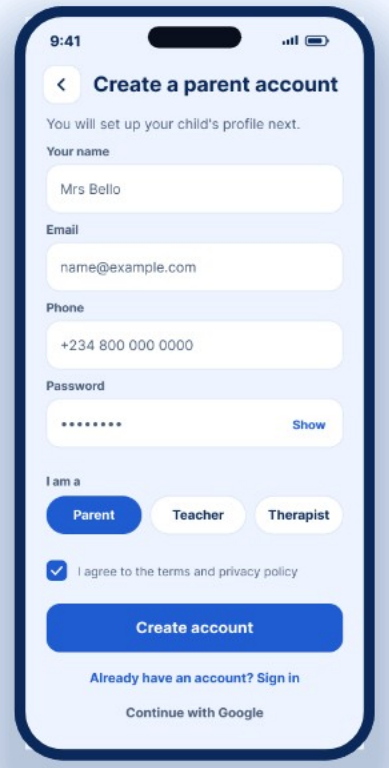
Progress & badges

Badges are collected, never lost. Progress is shown per subject as filled segments, the same visual language as the lesson step bar.

13 · KEY SCREENS — PARENTS, TEACHERS & SETTINGS

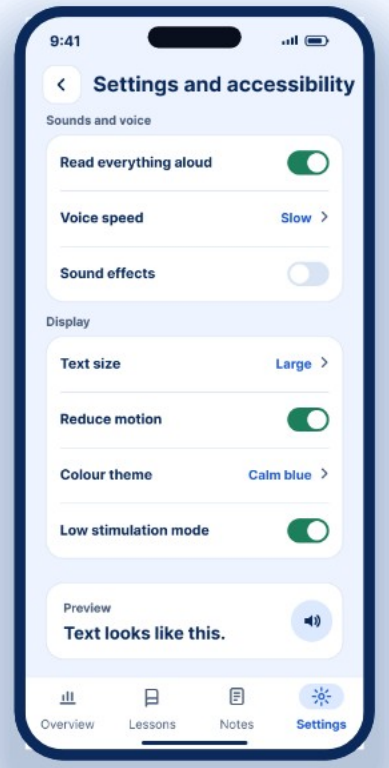
Control for adults, consistency for the learner

Adults set up the account, tune sound, text and motion, and follow progress. Nothing in these screens changes the learner's routine without the adult choosing it.



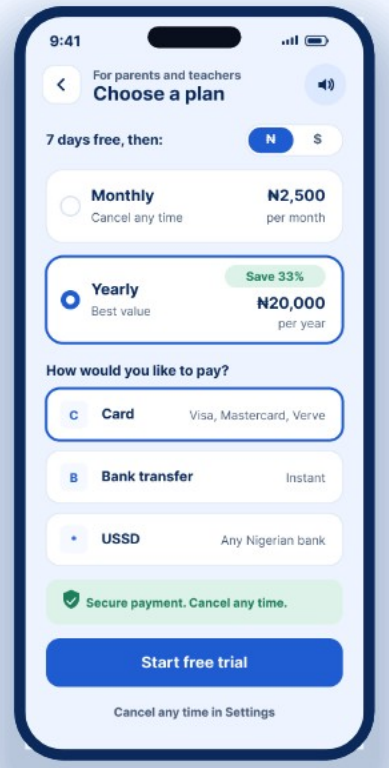
Sign up

One form with large fields and a role choice (parent, teacher, therapist). A single Google option sits below the main button, not beside it.



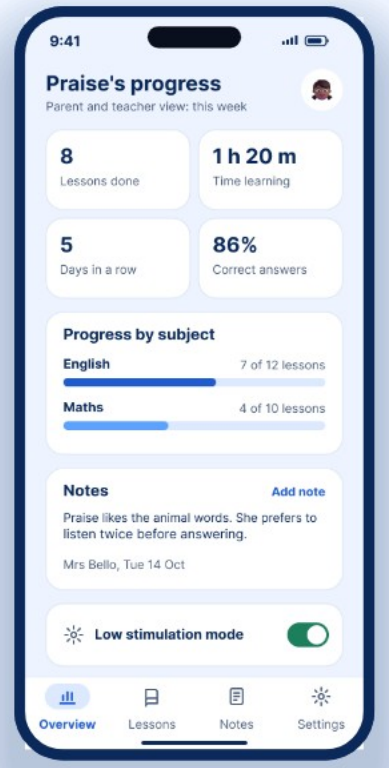
Settings & accessibility

Read everything aloud, voice speed, sound effects, text size, reduce motion, colour theme and a low-stimulation mode — plain switches with labels.



Subscription

Two plans priced in naira, a free trial first, and Nigerian payment options (card, bank transfer, USSD). The learner never sees this screen.



Parent / Teacher dashboard

Lessons done, time learning, correct answers, progress per subject and notes — the information Mrs Bello needs to plan the next session.

14 · ACCESSIBILITY

Checked against WCAG 2.2 and the autism-specific guidance, screen by screen

The checklist below is what the design was verified against at the end of the project. Values marked “measured” were calculated from the actual colour tokens and components in the Figma file.

AREA	WHAT THE DESIGN DOES	REFERENCE
✓ **Colour contrast**	All text on the palette: 4.6:1 to 12.7:1 (measured). Body text on Ice 12.7:1; muted text on white 6.7:1; white on Primary 6.0:1; white on the header gradient end 4.6:1.	WCAG 2.2 · 1.4.3 Contrast (Minimum) ≥ 4.5:1
✓ **Tap targets**	Primary buttons 56 px high and full width; listen pill 48 px; back button 44 × 44 px; quiz answers ≥ 160 px squares.	WCAG 2.2 · 2.5.8 Target Size (Minimum) ≥ 24 px; Apple HIG ≥ 44 pt
✓ **Motion**	No autoplaying animation. Lesson motion only on tap (block slides, 300–500 ms, ease-out). “Reduce motion” switch in Settings turns it off.	WCAG 2.2 · 2.2.2 Pause, Stop, Hide; 2.3.3 Animation from Interactions
✓ **Flashing**	Nothing flashes. Rewards are a static or slowly glowing star; no confetti, strobes or shaking.	WCAG 2.2 · 2.3.1 Three Flashes or Below Threshold
✓ **Audio**	Every instruction, word, sum and question has a listen button. Audio never starts on its own; a speaker icon marks what can be heard.	UDL 3.0 · multiple means of representation; W3C COGA · Clear language (written or audio)
✓ **Language**	Plain words, one short sentence per instruction, no idioms (“S says sss”, “Take 2 apples away”). Reading level is kept to the learner’s age or below.	W3C COGA · Use clear words; Use literal language; WCAG 3.1.5 Reading Level (AAA, aspirational)
✓ **Text size**	Minimum 14 px; body 16–18 px; lesson words 36–120 px. “Larger text” switch in Settings.	WCAG 2.2 · 1.4.4 Resize Text; 1.4.12 Text Spacing
✓ **Consistency & predictability**	Identical header, step bar, button positions and bottom navigation across all 30 screens; the same “I need a break” on every lesson.	WCAG 2.2 · 3.2.3 Consistent Navigation; 3.2.4 Consistent Identification; TEACCH
✓ **Error handling**	Wrong quiz answers are designed to fade and the question is re-read; there are no time limits anywhere in a lesson.	WCAG 2.2 · 2.2.1 Timing Adjustable; 3.3.1 Error Identification
… **Still to verify**	Screen-reader labels, focus order and VoiceOver behaviour can only be checked in a built prototype; colour-blind simulation on the blue-only palette passes by design but is listed as to-do.	WCAG 2.2 · 1.3.1 Info and Relationships; 4.1.2 Name, Role, Value

15 · OUTCOME & NEXT STEPS

What was delivered, and what happens before anyone calls it finished

This project is a complete design, not a tested product. The outcome is the design work itself; the measurements come after the planned test.

DELIVERED (OCTOBER 2026)

- 30 high-fidelity iOS screens (390 × 844) plus a cover, all editable in Figma
- English: topics, phonics, word + picture, build a sentence, picture story, quiz
- Maths: topics, counting, addition, subtraction, multiplication, division, brackets, fractions, quiz
- Routine and regulation: home, daily schedule, well done, calm break, progress & badges
- Adult side: onboarding, sign up, settings & accessibility, subscription, parent/teacher dashboard
- Design system: colour tokens, Inter type scale, buttons, cards, step bar, icon set
- Human-made imagery only: Microsoft Fluent 3D objects (MIT), Storyset scenes, real animal photographs (Wikimedia Commons), with a credits page

NEXT STEPS

1. **Usability test**
   Moderated sessions with 5–6 autistic learners and their adults, as described in the research section. Fix what breaks the routine before any development.
2. **Motion prototype**
   Build the on-tap lesson motion (blocks sliding together, letters lighting up) as a Figma prototype with reduced-motion variants, so it can be tested rather than described.
3. **Audio & language**
   Record the read-aloud lines with a clear, calm voice; explore Nigerian English pronunciation and a Yoruba/Hausa/Igbo option for instructions.
4. **Teacher content tools**
   Let teachers add their own word lists and photos to the lesson shell, keeping the layout fixed while the content becomes local.

A NOTE ON NUMBERS

This case study reports no completion rates, satisfaction scores or learning gains, because none have been measured yet. Every claim about autistic learners above is referenced to a published source in section 16; every claim about the app is visible in the screens.

Everything this case study leans on

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ILLUSTRATION & PHOTO CREDITS

- 3D objects and characters: Microsoft Fluent Emoji (3D style), Microsoft Design — MIT licence, github.com/microsoft/fluentui-emoji. Used as supplied, trimmed and composed; no generation.
- Scene illustrations (cover, splash, calm break): Storyset by Freepik, "Rafiki" style, recoloured to the app blue — "Illustration by Storyset", storyset.com.
- Cat: "Cat November 2010-1a.jpg" by Alvesgaspar, CC BY-SA 3.0 · Dog: "Labrador Retriever portrait.jpg" by Herwig Kavallar, public domain · Bird: "Erithacus rubecula profile.jpg" by C-M, CC BY-SA 4.0 · Fish: "Carassius auratus auratus (goldfish) 1.jpg" by James St. John, CC BY 2.0 — all via Wikimedia Commons; backgrounds removed.
- Case study layout, wireframes, diagrams and all text: Clement Olatunde. No AI-generated imagery is used anywhere in the app or in this case study.

StepWise

Thank you for reading

StepWise is a UX/UI case study by Clement Olatunde. Feedback from autistic learners, parents, teachers and designers is very welcome — especially before the usability test.

Clement Olatunde · UX/UI Designer

Designed in Figma · 2026

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